

Locating Voltage Sag Source with Impedance Measurement

Zhenguo Shao, JinPing Peng, Jian Kang

Abstract—Voltage sag is one of the major power quality problems that maybe result from networks fault and sudden increase of the load. It will result in abnormal operation or shutting down of device. Sag location is an important topic because power-quality trouble-shooting, diagnosis, mitigation policy could be constituted and mitigation measurement be carried out only after locating the voltage sag source accurately. A method based on the impedance measurement in $dq0$ coordinate system is proposed in this paper. Firstly, the three-phase voltage and current measured at the monitoring point are transformed by the orthogonal Park's transformation. And then the certain character of obtained three-phase instantaneous impedance measurement is used to locate the sources of the voltage sags. Constant impedance load and induction-motor load are used in digital simulation respectively. The simulation results show that the method can locate the disturbance source whether symmetrical or asymmetrical happens.

Index Terms—voltage sag source, location, impedance measurement, induction-motor

I. INTRODUCTION

VOLTAGE sag is one of the major power quality problems that maybe result from networks fault such as equipments struck by lightning. Sudden increase of the load, such as motor-starting, will cause voltage sags too [1]. As the sensitive equipments are used widely the transient sag of the supply voltage will result in abnormal operation or shutting down. Researches on voltage sags are focus on how to detect RMS voltage sags [2]-[4]. But sag location is another important topic [5]-[7]. When the voltage sag happens in power system, the location of the disturbance source should be detected quickly and accurately. Because only after locating the voltage sag source, can power-quality trouble-shooting, diagnosis, mitigation policy be constituted and mitigation measurement be carried out.

It seemed that the energy flow at the monitoring point increases under downstream faults and decreases under upstream faults. So the location method concerning disturbance energy was proposed in [8]. It was accomplished by examining the energy flow and peak instantaneous power for both capacitor energizing and voltage sag disturbances. If the

direction of power oscillates around zero, it can't determine which side of the monitoring point the voltage sag originates from. And that method could only apply to symmetrical voltage sags.

The direction of instantaneous active- and reactive- power and energy based on variation components defined in [9] were used to detect the voltage sag source. It was verified that this method could be applied to symmetrical and asymmetrical voltage sag source location.

Slope sign of the voltage-current characteristic was used to response the disturbance source location [10] based on the assumption that the current measured at the monitoring point increases with downside faults and decreases with upper-side faults. The sign of the active current was used to determine the voltage sag source location in [11]. It required less parameter for calculation and less computational time for data processing.

[12] and [13] were based on the fact that the equivalent impedance would change during the voltage sag. So, the real part of the evaluated impedance and the impedance amplitude and angle were used to detect the disturbance source respectively. But these methods cannot work well in the event of asymmetrical voltage sags [14].

The orthogonal Park's transformation [15] is used to compute the value of voltage and impedance in this paper. Then according to the changing situation of the waveforms, the voltage sag source location can be determined. Firstly, distinguish whether there is voltage sag origination through detecting RMS voltage. And if there is voltage sag, the change of measurement impedance is used to judge which side the voltage sag happens on. Constant impedance load and induction-motor load are used in digital simulation respectively. The simulation results show that the method can locate the disturbance source whether load is constant impedance or induction motor with symmetrical or asymmetrical faults.

II. MEASUREMENT IMPEDANCE OF A THREE-PHASE SYSTEM

The measurement impedance defined as the ratio of voltage to current at monitoring point changes both in magnitude and angle during the proceeding of voltage sag [12]. For a phase-a-to-ground fault, the magnitude and angle of the seen impedance at the monitoring point will be wrote as

$$|Z| = \frac{U_a}{I_a} \quad (1)$$

$$\angle Z = \angle \dot{U}_a - \angle \dot{I}_a \quad (2)$$

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where U_a is voltage amplitude of phase-a, I_a is current amplitude of phase-a, $\angle U_a$ and $\angle I_a$ are the angles of the voltage and current phasor respectively.

The impedance measured in [12] was computed by the phasors of voltage and current. It maybe results in completely different conclusion as the voltage sag is a transient power quality event. Moreover, when the asymmetrical voltage sag is provoked, it is required to obtain voltage and current phasors per phase.

Signing the voltage and current vectors of the three-phase system as $u(t)$ and $i(t)$ respectively.

$$u(t) = \begin{bmatrix} u_a(t) \\ u_b(t) \\ u_c(t) \end{bmatrix} \quad i(t) = \begin{bmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{bmatrix} \quad (3)$$

Then the length of the vector $u(t)$, $\|u(t)\|$, could be computed in (4) where $[,]$ represent the inner product of vectors.

$$\begin{aligned} \|u(t)\| &= \sqrt{[u(t), u(t)]} = \sqrt{u(t)^T u(t)} \\ &= \sqrt{u_a^2(t) + u_b^2(t) + u_c^2(t)} \end{aligned} \quad (4)$$

According to the definition of instantaneous power in (5) the instantaneous impedance measurement of a three phase system could be computed as described in (7).

$$\begin{aligned} p(t) &= [u(t), i(t)] = u(t)^T i(t) \\ &= u_a(t)i_a(t) + u_b(t)i_b(t) + u_c(t)i_c(t) \end{aligned} \quad (5)$$

$$i(t) = \frac{u(t)^T i(t)}{u(t)^T u(t)} u(t) = \frac{p(t)}{\|u(t)\|^2} u(t) \quad (6)$$

$$\begin{aligned} z(t) &= \frac{u(t)}{i(t)} = \frac{\|u(t)\|^2}{p(t)} = \frac{\sqrt{[u(t), u(t)]}^2}{[u(t), i(t)]} \\ &= \frac{u_a^2(t) + u_b^2(t) + u_c^2(t)}{u_a(t)i_a(t) + u_b(t)i_b(t) + u_c(t)i_c(t)} \end{aligned} \quad (7)$$

Park's transformation is a mathematical transformation used to simplify the analysis of three-phase circuits as described in (8) and (9) where P denotes the orthogonal Park's transformation matrix.

$$\begin{bmatrix} u_d \\ u_q \\ u_0 \end{bmatrix} = P \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} \quad (8)$$

$$P = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \omega t & \cos(\omega t - \frac{2\pi}{3}) & \cos(\omega t + \frac{2\pi}{3}) \\ -\sin \omega t & -\sin(\omega t - \frac{2\pi}{3}) & -\sin(\omega t + \frac{2\pi}{3}) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad (9)$$

The Park's transformation matrix P is orthogonal, so $P^{-1} = P^T$ and $P^T P = E$.

$$\begin{aligned} [u_{dq0}, u_{dq0}] &= [P u_{abc}, P u_{abc}] \\ &= u_{abc}^T P^T P u_{abc} = u_{abc}^T u_{abc} \\ &= \|u(t)\| \end{aligned} \quad (10)$$

$$\begin{aligned} [u_{dq0}, i_{dq0}] &= [P u_{abc}, P i_{abc}] \\ &= u_{abc}^T P^T P i_{abc} = u_{abc}^T i_{abc} \\ &= p(t) \end{aligned} \quad (11)$$

Equation 10 and Equation 11 imply that the instantaneous impedance could also be computed in $dq0$ coordinate system.

Because the voltage sag is associated with harmonic components, it is necessary to extract the DC quantities through the low-pass filter after the Park's transformation. In this paper, the Butterworth low-pass filter is applied. The impedance measurement is expressed as

$$\begin{aligned} z(t) &= \frac{\|u(t)\|^2}{p(t)} \\ &= \frac{u_d^2(t) + u_q^2(t) + u_0^2(t)}{u_d(t)i_d(t) + u_q(t)i_q(t) + u_0(t)i_0(t)} \end{aligned} \quad (12)$$

III. THE EFFECT OF INDUCTION-MOTORS

A simple distributed system is shown in Fig.1 where voltage sags might originate either at F_1 or at F_2 . Z_{l1a} , Z_{l1b} , Z_{l2a} , and Z_{l2b} are the transmission line impedances. When the voltage sag source originates from the left side of the monitor (fault at F_1), the impedance measurement at the monitoring point is equal to that of before the voltage sag happens ($Z(t)_{sag} = Z_{l2a} + Z_{l2b} + Z_L = Z(t)_{presag}$). When the disturbance originates from the right side of the monitor (fault at F_2), the impedance measurement is $Z(t) = Z_{l2a} + R_g / (Z_{l2b} + Z_L)$ and $Z(t)_{sag} < Z(t)_{presag}$. The conclusion obtained above is just for systems with constant impedance loads. If there are induction-motors fed in that system, the transient of motor will change the value of impedance measurement.

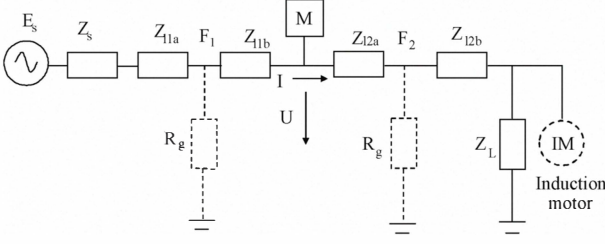


Fig. 1. Equivalent circuit for voltage sag analysis

As described in Fig.2 the equations of the stator circuits could be written as follows where r_s is the stator resistance.

$$u_{sd} = r_s i_{sd} + p\psi_{sd} - \omega_s \psi_{sq} \quad (13)$$

$$u_{sq} = r_s i_{sq} + p\psi_{sq} - \omega_s \psi_{sd} \quad (14)$$

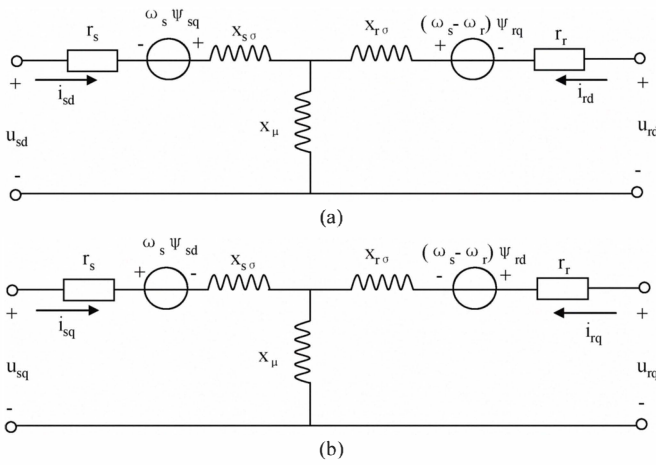


Fig. 2. Mathematical models for Induction-motor electromagnetic transient. (a) d-axis equivalent circuit. (b) q-axis equivalent circuit.

The magnetic circuit of the induction-motor is symmetrical, that is, the d-axis parameters are the same as q-axis parameters. Therefore, the above equation can be rewritten into plural form.

$$\vec{U}_s = r_s \vec{I}_s + p\vec{\Psi}_s + j\omega_s \vec{\Psi}_s \quad (15)$$

Neglecting the stator electromagnetic transient and the variation of ω_s , (15) is simplified to (16).

$$\vec{U}_s = r_s \vec{I}_s + j\vec{\Psi}_s \quad (16)$$

The windings flux of stator and rotor can be written in (17) and (18) respectively where $X_s = X_\mu + X_{s\sigma}$, $X_r = X_\mu + X_{r\sigma}$.

$$\vec{\Psi}_s = X_s \vec{I}_s + X_\mu \vec{I}_r \quad (17)$$

$$\vec{\Psi}_r = X_\mu \vec{I}_s + X_r \vec{I}_r \quad (18)$$

Solving $\vec{\Psi}_s$ and $\vec{I}_r$ from (17) and (18) and substituting into (15)

$$\begin{aligned} \vec{U}_s &= (r_s + jX') \vec{I}_s + \vec{E}' \\ X' &= X_s - \frac{X_\mu^2}{X_r} \\ \vec{E}' &= j \frac{X_\mu}{X_r} \vec{\Psi}_r \end{aligned} \quad (19)$$

At the beginning of the voltage sag, the motor sub-transient EMF, $\vec{E}'$, remains unchanged. The terminal voltage of the motor will be lower than $\vec{E}'$. Therefore, the motor become an additional power source temporarily and supply part of the short-circuit current. Its supply of short-circuit sub-transient current is $(\vec{E}' - \vec{U}_s) / (r_s + jX')$. In order to maintain the flux conservation at the moment of short trouble, the stator windings also produce non-periodic current. Because the resistance of the induction-motor is higher than the reactance, the periodic and non-periodic currents, supplied by the motor, are rapidly fading. The decay time of the two components is approximated about a few percent seconds.

The impedance measurement at the monitoring point is

$$\begin{aligned} z(t) &= \frac{\|u(t)\|^2}{p(t)} = \frac{u_d^2 + u_q^2}{u_d i_d + u_q i_q} \\ &= \frac{(u_{sd} + i_{sd} \cdot z_{l2})^2 + (u_{sq} + i_{sq} \cdot z_{l2})^2}{(u_{sd} + i_{sd} \cdot z_{l2}) i_{sd} + (u_{sq} + i_{sq} \cdot z_{l2}) i_{sq}} \end{aligned} \quad (20)$$

Where $z_{l2} = Z_{l2a} + Z_{l2b}$. It's invariant when the voltage is constant and could be written as (21). If the voltage sag originates from the left side of the monitor, $z(t)$ may be less than zero at first and then approach to the value less than Z_{steady} .

$$Z_{steady} = r_s + jx_{s\sigma} + (r_\mu + jx_\mu) / ((r_r / s + jx_{r\sigma}) + z_{l2}) \quad (21)$$

When the disturbance source originates from the right side of the monitor, the impedance measurement at the monitoring point is computed in (22), where $u_{sd} = u_{sd} + i_{sd} Z_{l2b}$ and $u_{sq} = u_{sq} + i_{sq} Z_{l2b}$.

$$\begin{aligned} z_{sag} &= \frac{\|u(t)\|^2}{p(t)} = \frac{u_d^2 + u_q^2}{u_d i_d + u_q i_q} \\ &= \frac{u_{sd}^2 + u_{sq}^2}{\frac{u_{sd}^2 + u_{sq}^2}{R_g} + u_{sd} i_{sd} + u_{sq} i_{sq}} + z_{l2a} \end{aligned} \quad (22)$$

In conclusion, if $0 < Z_{sag} < Z_{steady}$ during the voltage sag, the disturbance source originates from the right side of the monitoring point. Otherwise, the voltage sag source is on the left side of the monitoring point.

IV. SIMULATION OF SAG LOCATION

MATLAB/SIMULINK was used to simulate the location of sag provoked by 3-phase short-circuit fault, phase-to-phase short-circuit fault and phase-to-ground fault in 400V system.

A. Power system with constant impedance

As shown in Fig.3, the three-phase faults on both side of the monitoring device were simulated as voltage sag sources from different sides (the fault resistance R_{on} was set to 2Ω). The left side 3-phase fault last from 0.3 to 0.4 seconds and then the right side fault last from 0.7 to 0.8 seconds.

For different sides and different types of voltage sags, the RMS voltages and impedance measurements at the monitoring point are plotted in Fig.4 (a), (b) and (c). It can be seen that the impedance measurements almost remain unchanged during the first voltage sag. It indicates that the disturbance source is on the left side of the monitoring point (fault at F_1). In the second sag the impedance measurements dip but are still greater than zero in 0.7-0.8 seconds. It indicates that the disturbance source is on the right side of the monitoring point (fault at F_2).

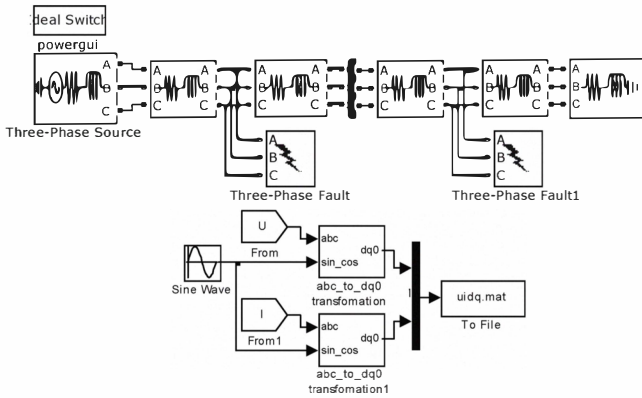


Fig. 3. Power system with constant impedance load

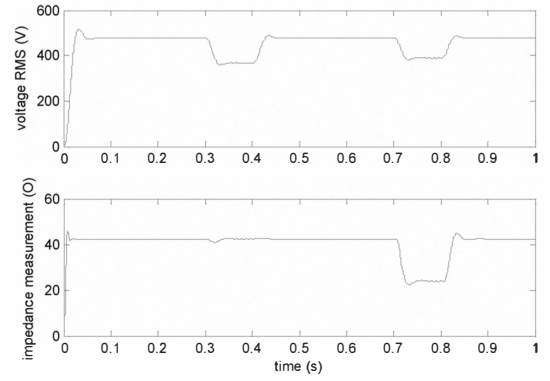
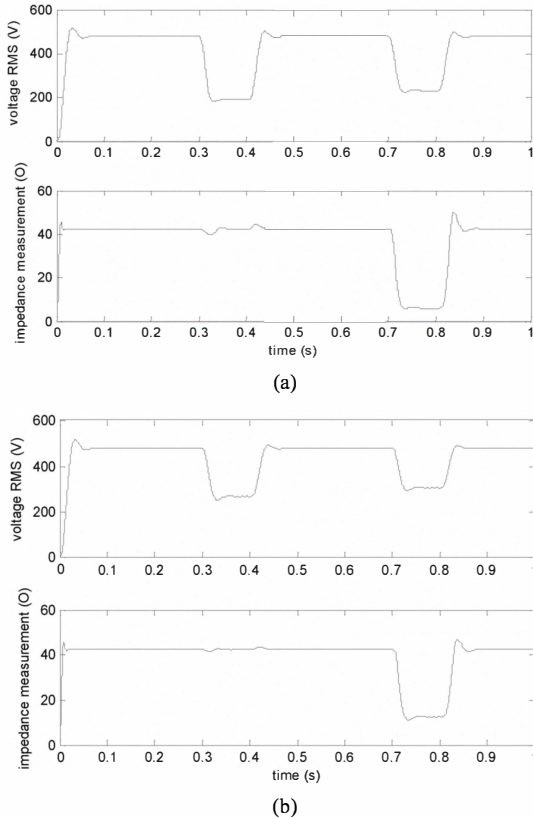


Fig. 4. Simulation results for constant impedance load. (a) Voltage sags caused by 3-phase faults at F_1 (0.3-0.4s) and F_2 (0.7-0.8s) respectively. (b) Voltage sags caused by inter-phase faults at F_1 (0.3-0.4s) and F_2 (0.7-0.8s) respectively. (c) Voltage sags caused by phase-a grounding faults at F_1 (0.3-0.4s) and F_2 (0.7-0.8s) respectively.

B. Power system with induction-motor

Motor has an important effect on the voltage characteristics during a three-phase fault condition. It was verified with the model shown in Fig. 5. The asynchronous motor parameters were set to 10HP (7.5KW), 400V, 50Hz and 1440RPM. The faults on both side of the monitoring device were simulated as voltage sag sources respectively. The fault resistance R_{on} was set to 1Ω . The faults included three-phase fault, inter-phase fault and grounding fault. The left side three-phase fault last from 0.1 to 0.15 seconds and then the right side fault last from 0.6 to 0.65 seconds.

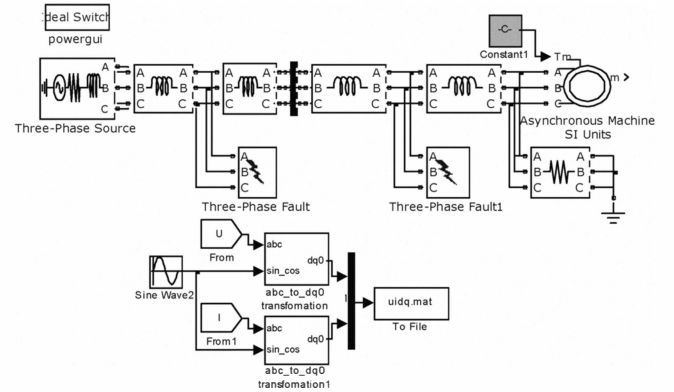


Fig. 5. Power system with induction-motor.

The impedance measurements at the monitoring point for different types of voltage sags caused by short-circuit faults are plotted in Fig.6 (a), (b) and (c). It can be seen that the impedance measurements are negative at the beginning of the first voltage sags. Subsequently the impedance measurements become positive but less than the values before voltage sags indicating that the disturbance sources are on the left side of the monitoring point (fault at F_1). At the beginning of the first voltage sags, the motor becomes a temporary power and releases energy. Therefore, the instantaneous active power $p(t)$ at the monitoring point is negative and so the measured impedance $z(t)$ is negative too. About a few milliseconds later, the current supplied by the motor decays to zero and the

terminal voltage of motor swells, so the measured impedance $z(t)$ increases to the value just little less than the values before voltage sags. In the second sag the measured impedance dip but still greater than zero. It indicates that the disturbance sources are on the right side of the monitoring point (fault at F_2). Because the system on the right side of the monitor could be equivalent to the fault resistance parallel connected to motor, the measured impedance at the monitoring point dips and is still greater than zero.

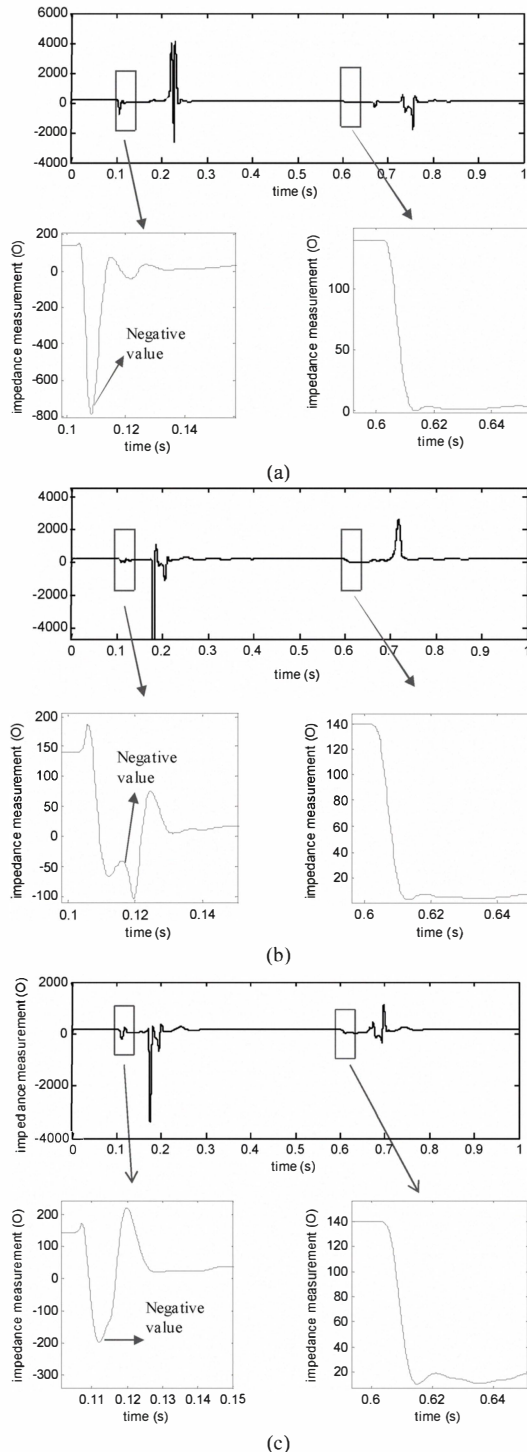


Fig. 6. Simulation results for induction-motor. (a) Voltage sags caused by 3-phase faults at F_1 (0.1-0.15s) and F_2 (0.6-0.65s) respectively. (b) Voltage sags

caused by inter-phase faults at F_1 (0.1-0.15s) and F_2 (0.6-0.65s) respectively. (c) Voltage sags caused by phase-a grounding faults at F_1 (0.1-0.15s) and F_2 (0.6-0.65s) respectively.

V. CONCLUSION

The method proposed in this paper applies the orthogonal Park's transformation to obtain the amplitude of the three-phase voltages and currents in the d-q coordinate system. And then extract the certain character of measured impedance to locate the voltage sag source. There is no need to compute voltage and current phasors per phase of the fundamental frequency when voltage sags are provoked by asymmetrical earth fault. Furthermore, the effect of the induction-motor on the sag location was discussed. At the beginning of the first voltage sags, the motor becomes a temporary power and releases energy to fault resistance that could change the sign of measured impedance.

The method for voltage sag source detection was tested with MATLAB/SIMULINK in the cases of constant load and induction-motor. And it was effective not only in symmetrical voltage sags but also in asymmetrical voltage sags.

VI. REFERENCES

- [1] M. J. H. Bollen, "Voltage sags in three-phase systems," *IEEE Power Engineering Review*, vol. 21, no. 9, pp. 8-11,15, September 2001.
- [2] R. Naidoo, P. Pillay, "A New Method of Voltage Sag and Swell Detection," *IEEE transactions on power delivery*, vol. 22, no. 2, April 2007.
- [3] A. Florio, A. Mariscotti, M. Mazzucchelli, "Voltage Sag Detection Based on Rectified Voltage Processing," *IEEE transactions on power delivery*, vol. 19, no. 4, October 2004.
- [4] G. Lv, X. D. Wang, "Voltage Sags Detection And Identification Based On Phase-Shift And RBF Neural Network," *Department of Information Science and Engineering, Zhejiang Normal University, Jinhua, China*.
- [5] J. E. Jipping and W. E. Carter, "Application and experience with a 15kV static transfer switch," *IEEE Transactions on Power Delivery*, vol. 14, no. 4, pp. 1477-1481, October 1999.
- [6] E. Sttvaktakis, I. Y. H. Gu and M. J. H. Bollen, "Voltage dip detection and power system transients," in *Proceedings of Power Engineering Society Summer Meeting*, vol. 1, pp. 683-688, July 2001.
- [7] C. Fitzer, M. Barnes and P. Green, "Voltage sag detection technique for a dynamic voltage restorer," *IEEE transactions on industry applications*, vol. 40, no. 1, pp. 203-212, January 2004.
- [8] A.C. Parsons, W. M. Grady, E. J. Powers, and J. C. Soward, "A Direction Finder for Power Quality Disturbances Based upon Disturbance Power and Energy," *IEEE Transactions on Power Delivery*, vol. 15, no.3, pp.1081-1086, July 2000.
- [9] W. Kong, X. Z. Dong, and Z. Chen, "Voltage Sag Source Location Based on Instantaneous Energy Detection," *Electric Power Systems Research*, 78 (2008) 1889-1898.
- [10] C. Li, T. Tayjasanant, W. Xu, and X. Liu, "Method for Voltage-sag-source Detection by Investigating Slope of the System Trajectory," *IEE Proceedings: Generation, Transmission and Distribution*, vol. 150, no.3, pp. 367 - 372, May 2003.
- [11] N. Hamzah, A. Mohamed, and A. Hussain, "A New Approach to Locate the Voltage Sag Source Using Real Current Component," *Electric Power Systems Research*, vol.72, no.2, pp.113 - 123, December 2004.
- [12] A. K. Pradhan and A. Routray, "Applying Distance Relay for Voltage Sag Source Detection," *IEEE Transactions on Power Delivery*, vol.20, no.1, pp.529-531, January 2005.
- [13] T. Tayjasanant, V. Li, and W. Xu, "A Resistance Sign-based Method for Voltage Sag Source Detection," *IEEE Transactions on Power Delivery*, vol.20, no.4, pp.2544-2551, October 2005.
- [14] R.-C. Leborgne, D. Karlsson, and J. Daalder, "Voltage sag source location methods performance under symmetrical and asymmetrical fault

conditions," in *Proceedings of transmission and distribution conference and exposition*, Latin America, August 2006.

- [15] Y. X. Ni, S. S. Chen, B. L. Zhang, "*Theory and Analysis of Dynamic Power System*". Beijing: Tsinghua University Press, 2002, pp. 14.